Two-Stage Gait and Posture Analysis Methodology

for the RMD Wearable IMU Dataset

Classical gait characterization, lower-limb multivariate functional PCA, whole-body transfer entropy, and spectral clustering

Purpose. This document is an implementation-ready technical reference for the planned gait/posture study. It is written to serve two audiences simultaneously: (i) the research team preparing a journal manuscript, and (ii) a coding agent such as Codex implementing the full analysis pipeline. Mathematical definitions, analysis decisions, expected data products, quality-control rules, and software interfaces are therefore made explicit.

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1 Executive summary and study architecture

The proposed study is deliberately modular. Stage 1 is designed as a scientifically interpretable dataset-characterization layer that can stand on its own if the manuscript is pursued as a *Scientific Data* Data Descriptor. Stage 2 adds a systems-level coordination and unsupervised-pattern-discovery layer that is intended to provide the additional hypothesis-driven contribution needed for a *Scientific Reports* Article.

The analysis is organized around four progressively richer descriptions of gait:

1. **Classic gait features:** what changed in timing and joint kinematics?
2. **Lower-limb functional modes:** what dominant multijoint waveform patterns explain the changes?
3. **Whole-body directed information transfer:** how did the statistical coupling structure of body motion reorganize?
4. **Spectral clustering of trial phenotypes:** do recurrent movement/adaptation patterns emerge across participants, terrain, and footwear?

The resulting logic is

$$\boxed{\begin{aligned}&\text{IMU data} \rightarrow \text{gait events and phase} \rightarrow \text{classic gait} \rightarrow \text{lower-limb MFPCA}\\ &\qquad\qquad \rightarrow \text{whole-body TE network} \rightarrow \text{trial phenotype} \rightarrow \text{spectral clustering.}\end{aligned}} \tag{1}$$

Primary scientific framing

The central contribution should not be framed as a population-normative gait study. The current repository contains many gait cycles but only seven independent participants. The strongest defensible framing is therefore **within-participant adaptation and emergence of recurring coordination strategies under terrain and footwear perturbations**, with participant-level uncertainty and no cycle-level pseudoreplication.

2 Source basis and non-negotiable data assumptions

This methodology consolidates the supplied RMD dataset description, the submitted Hilbert-phase gait coordination study, the submitted lower-limb transfer-entropy study, and the earlier feasibility assessment. The dataset description reports seven gait participants, 220 recording sessions, 345 gait/posture CSV files, approximately 179 minutes of gait/posture data sampled at 100 Hz, and 208 currently processed gait recordings containing 3,343 labeled gait cycles [1]. The current processing tree maps 18 recorded sensor channels to body segments and contains additional raw IMU channels that are not used by the present segment mapping [1].

The supplied Hilbert study defines an analytic-signal gait phase and uses instantaneous phase, continuous relative phase, phase-locking value, phase variability, and cyclograms to characterize lower-limb coordination across terrain [2]. The supplied transfer-entropy study uses pairwise directed transfer entropy among hip, knee, and ankle signals, followed by graph-derived inflow, outflow, and net-flow quantities [3]. The proposed work extends those concepts but changes their scope substantially.

2.1 Important processing-pipeline discrepancy to resolve before coding

The supplied documents do not describe exactly the same joint-angle extraction path. The previous submitted gait papers describe Kalman-filter-based denoising followed by PCA or wavelet-PCA

processing before joint-angle analysis [2, 3]. By contrast, the current RMD dataset description explicitly distinguishes the platform manuscript's validated joint-angle estimator from the repository's RSS research chain and warns that RSS rotational modes are not validated anatomical joint angles [1].

Required team decision before final analysis

The new journal analysis must use **one locked, anatomically interpretable joint-angle pathway**. This document assumes the validated segment-orientation / proximal-to-distal relative-rotation pathway described in the RMD dataset note. If the team intentionally chooses the earlier EKF/UKF+PCA pathway instead, that choice must be documented and applied consistently. Do not combine anatomical joint angles from one pipeline with RSS mode trajectories from another and label them as the same quantity.

The RMD note also states that the current exported segment-quaternion files are empty apart from indexing/labels. Any analysis requiring gravity-referenced absolute segment posture must therefore re-run or repair the validated orientation export rather than infer posture from RSS modes [1].

3 Notation, hierarchy, and statistical unit

Let

- $s \in \{1, \ldots, S\}$ index participants,
- c index experimental condition (terrain or footwear),
- r index repeated trial/session within a participant-condition pair,
- i index gait cycles within a trial,
- j index biomechanical signal/node,
- t denote continuous time or sample index,
- $\tau \in [0, 1]$ denote normalized within-stride phase.

A raw time series is denoted $x_{scrj}(t)$. A cycle-normalized trajectory is denoted $X_{scrij}(\tau)$.

The data hierarchy is

$$\text{participant} \supset \text{condition} \supset \text{trial} \supset \text{gait cycle}. \tag{2}$$

Cycles are repeated measurements used to estimate a trial or participant-condition response. They are *not* independent subjects. Therefore, the default inferential and clustering unit is the **trial**. Participant-level resampling is used for uncertainty whenever conclusions are generalized beyond a single participant.

4 End-to-end workflow

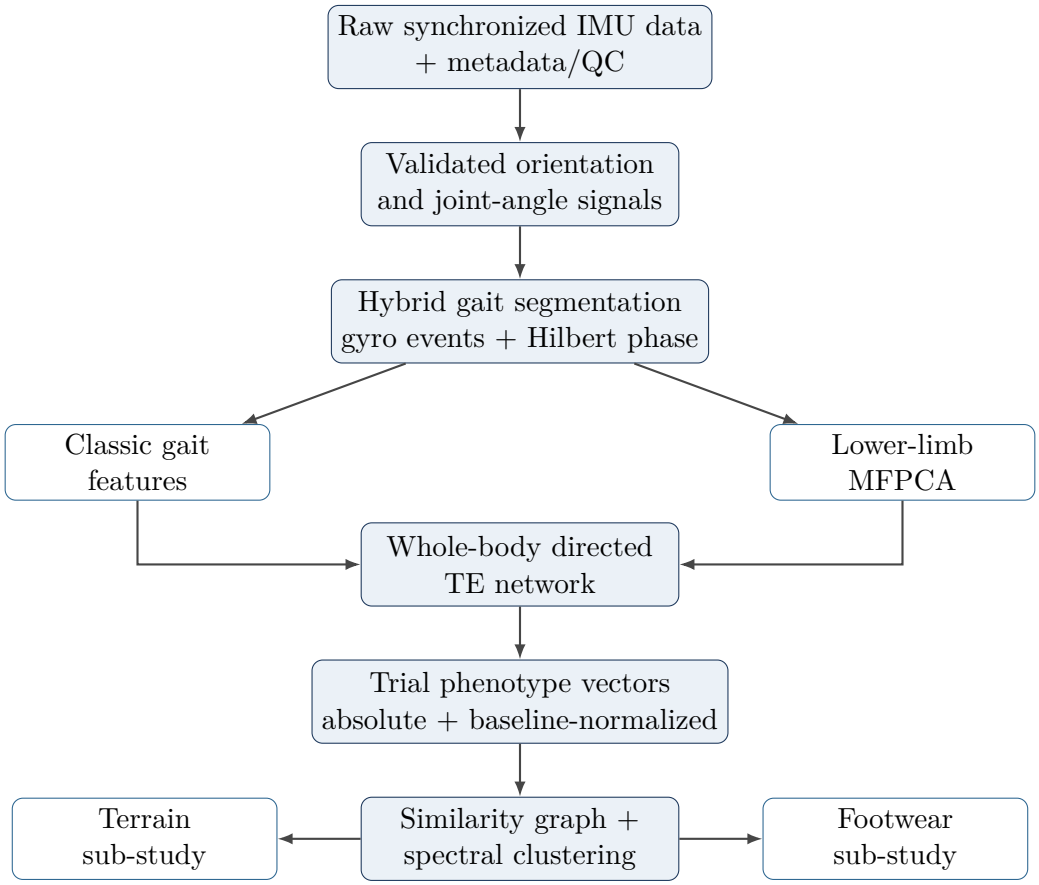


Figure 1: Recommended modular workflow. Stage 1 consists of segmentation, classic gait characterization, and lower-limb multivariate functional PCA. Stage 2 adds whole-body transfer-entropy networks and spectral clustering.

5 Cohort locking, metadata, and preprocessing

5.1 Build a participant-condition-trial manifest first

Before outcome analysis, create a frozen manifest with one row per source recording. At minimum it should contain

$$\mathcal{M} = \{\texttt{participant}, \texttt{terrain}, \texttt{direction}, \texttt{footwear}, \texttt{heel_code}, \texttt{trial}, \texttt{file}, \texttt{duration}, \texttt{sensor_availability}, \texttt{qc_status}\}. \tag{3}$$

The manifest must normalize inconsistent folder naming and record known sensor dropout, truncated recordings, missing channels, and whether a valid periodic segment was found. No record should disappear silently from later stages.

5.2 Separate terrain and footwear sub-studies

The condition matrix is unbalanced. Consequently, terrain and footwear effects should not be estimated from one undifferentiated pool unless a sufficiently balanced factorial subset exists.

Primary matched analysis subsets

Terrain sub-study: restrict to a reference footwear condition (preferably the flat/reference shoe code) and compare flat, slope ascent/descent, stair ascent/descent, and rough terrain only where participants share the required conditions.
Footwear sub-study: restrict to flat terrain and compare the reference footwear against repeated heel/platform conditions within the same participants.
Joint terrain-by-footwear analysis: exploratory only unless the manifest reveals a sufficiently complete repeated-measures factorial design.

5.3 Signal preprocessing

For each IMU channel, use synchronized acceleration $\boldsymbol{a}(t) = [a_x, a_y, a_z]^\top$ and angular velocity $\boldsymbol{\omega}(t) = [\omega_x, \omega_y, \omega_z]^\top$ at sampling rate $f_s = 100$ Hz. Processing should be zero-phase whenever offline analysis is intended so that event timing is not shifted.

A generic nth-order Butterworth band-pass is denoted

$$\tilde{x}(t) = \text{filtfilt}\Big(B_{n,[f_L,f_H]}, x(t)\Big)\,. \tag{4}$$

The earlier Hilbert study used gait-relevant filtering around 0.5–6 Hz [2]. This is an appropriate starting point for phase extraction, but filter cutoffs should be configurable and verified by sensitivity analysis rather than hard-coded as universal constants.

6 Stage 1A: hybrid gait segmentation

6.1 Biomechanical event definitions

For limb $q \in \{L, R\}$, define initial contact $\text{IC}_q^{(i)}$ and toe-off $\text{TO}_q^{(i)}$. A stride is

$$\mathcal{S}_q^{(i)} = [\text{IC}_q^{(i)}, \text{IC}_q^{(i+1)}), \tag{5}$$

with duration

$$T_{\text{stride},q}^{(i)} = \text{IC}_q^{(i+1)} - \text{IC}_q^{(i)}. \tag{6}$$

Stance and swing durations are

$$T_{\text{stance},q}^{(i)} = \text{TO}_q^{(i)} - \text{IC}_q^{(i)}, \tag{7}$$

$$T_{\text{swing},q}^{(i)} = \text{IC}_q^{(i+1)} - \text{TO}_q^{(i)}. \tag{8}$$

The stance fraction is

$$\rho_{\text{stance},q}^{(i)} = \frac{T_{\text{stance},q}^{(i)}}{T_{\text{stride},q}^{(i)}}. \tag{9}$$

A step time is defined between contralateral initial contacts, e.g.

$$T_{L\to R}^{(i)} = \text{IC}_R^{(k)} - \text{IC}_L^{(i)}, \tag{10}$$

where $\text{IC}_R^{(k)}$ is the first right initial contact after $\text{IC}_L^{(i)}$.

6.2 Gyroscope-based event detection

Shank sagittal-plane angular velocity is a standard signal for identifying mid-swing, initial contact, and foot-off events. Rule-based gyroscope methods have been validated on level and inclined walking and use the characteristic swing peak and neighboring valleys/zero crossings [5, 6]. In the RMD platform, the preferred event sensor is the distal foot IMU when its anatomical axis is reliable; the shank IMU is the fallback and cross-check.

If the sagittal gyroscope axis is known, denote it $g_q(t)$. If orientation conventions vary, construct a robust one-dimensional walking-plane signal from the three gyroscope axes using PCA over the periodic interval:

$$\boldsymbol{\Sigma}_\omega = \frac{1}{N-1}\sum_t (\boldsymbol{\omega}_t - \bar{\boldsymbol{\omega}})(\boldsymbol{\omega}_t - \bar{\boldsymbol{\omega}})^\top, \tag{11}$$

$$\boldsymbol{v}_1 = \arg\max_{\|\boldsymbol{v}\|=1} \boldsymbol{v}^\top \boldsymbol{\Sigma}_\omega \boldsymbol{v}, \qquad g_q(t) = \boldsymbol{v}_1^\top \boldsymbol{\omega}(t). \tag{12}$$

The sign of g_q is flipped if necessary so that the dominant mid-swing peaks are positive.

Candidate mid-swing events are local maxima satisfying

$$g_q(t) > \text{median}(g_q) + \alpha\,\text{MAD}(g_q), \tag{13}$$

with a minimum peak spacing corresponding to a plausible stride/step interval. α is configuration-controlled and should be tuned from development data, not from test outcomes.

For each candidate $\text{MS}_q^{(i)}$, search backward for $\text{TO}_q^{(i)}$ and forward for $\text{IC}_q^{(i)}$ using the nearest physiologically plausible valley/zero-crossing structure. For shank angular velocity, the pre-mid-swing valley approximates terminal contact and the post-mid-swing valley approximates initial contact in common rule-based formulations [5, 6]. For foot-mounted signals, a stationary-score transition may be used as an auxiliary check.

Define foot angular-speed magnitude

$$\Omega_q(t) = \|\boldsymbol{\omega}_q(t)\|_2 \tag{14}$$

and gravity-consistency residual

$$A_q(t) = \left|\|\boldsymbol{a}_q(t)\|_2 - g\right|. \tag{15}$$

A soft stationarity score can be implemented as

$$S_q(t) = \sigma\left(\frac{\theta_\omega - \Omega_q(t)}{s_\omega}\right)\sigma\left(\frac{\theta_a - A_q(t)}{s_a}\right), \qquad \sigma(u) = \frac{1}{1+e^{-u}}. \tag{16}$$

This score is not itself the final IC/TO estimator; it is used to reject implausible candidate events and to refine event timing when a foot IMU is available.

6.3 Hilbert analytic phase

For a filtered periodic reference signal $x(t)$, the Hilbert transform is

$$\hat{x}(t) = \mathcal{H}\{x(t)\} = \frac{1}{\pi}\,\text{P.V.}\int_{-\infty}^{\infty}\frac{x(\tau)}{t-\tau}\,d\tau. \tag{17}$$

The analytic signal is

$$z(t) = x(t) + j\hat{x}(t) = A(t)e^{j\phi(t)}, \tag{18}$$

with

$$A(t) = |z(t)| = \sqrt{x^2(t) + \hat{x}^2(t)}, \tag{19}$$

$$\phi(t) = \arg z(t), \tag{20}$$

$$\phi_u(t) = \text{unwrap}(\phi(t)). \tag{21}$$

A continuous phase variable is

$$\Phi(t) = \frac{\phi(t) \bmod 2\pi}{2\pi} \in [0, 1), \tag{22}$$

consistent with the previous Hilbert-based gait framework [2].

The new pipeline uses Hilbert phase as a **continuous cyclic coordinate and consistency check**, while biomechanical event timing is anchored by the foot/shank gyroscope detector. It therefore avoids assuming that a fixed Hilbert phase value necessarily equals toe-off across all terrains and footwear conditions.

6.4 Event fusion

For each stride candidate:

1. Detect MS, TO, and IC independently on left and right distal-limb gyroscope signals.
2. Compute Hilbert phase on a gait-dominant reference signal.
3. Require the event order $\mathrm{IC}_i < \mathrm{TO}_i < \mathrm{MS}_i < \mathrm{IC}_{i+1}$ (or the equivalent order produced by the chosen signal convention).
4. Reject duplicate events and implausible stride durations.
5. Compare Hilbert phase advance over the candidate stride. A valid stride should contain approximately one full phase revolution.
6. Assign an event confidence score based on gyro prominence, Hilbert-cycle consistency, bilateral alternation, and sensor-quality flags.

A convenient phase-consistency metric is

$$\epsilon_{\phi}^{(i)} = \left| \frac{\phi_u(\mathrm{IC}_{i+1}) - \phi_u(\mathrm{IC}_i)}{2\pi} - 1 \right|. \tag{23}$$

A stride may be rejected if $\epsilon_{\phi}^{(i)}$ exceeds a configurable tolerance.

6.5 Cycle normalization

For standard trajectory plots, normalize each accepted stride as

$$\tau = \frac{t - \mathrm{IC}_i}{\mathrm{IC}_{i+1} - \mathrm{IC}_i}, \qquad \tau \in [0, 1]. \tag{24}$$

Resample to a fixed grid, e.g. $\tau_m = m/100$, $m = 0, \ldots, 100$.

Because terrain and footwear can change stance/swing timing, two registrations should be retained:

- **Timing-preserving whole-stride normalization:** one linear mapping from IC_i to IC_{i+1}. Toe-off remains at its measured normalized location $\tau_{\mathrm{TO}}^{(i)}$.
- **Event-registered shape normalization:** map $\mathrm{IC}_i \to \mathrm{TO}_i$ to a fixed stance interval and $\mathrm{TO}_i \to \mathrm{IC}_{i+1}$ to a fixed swing interval. This removes gross stance/swing timing variation so MFPCA emphasizes waveform shape. The true stance fraction ρ_{stance} is retained separately as a classic gait feature.

The primary MFPCA should use event-registered trajectories, with timing-preserving MFPCA as a sensitivity analysis.

7 Stage 1B: classic gait analysis

Classic gait features are calculated at cycle level and then summarized at trial level. Trial summaries use robust estimators such as the median and median absolute deviation unless a distribution is approximately symmetric.

7.1 Temporal features

For each limb and cycle:

$$T_{\text{stride}} = \text{IC}_{i+1} - \text{IC}_i, \tag{25}$$
$$T_{\text{stance}} = \text{TO}_i - \text{IC}_i, \tag{26}$$
$$T_{\text{swing}} = \text{IC}_{i+1} - \text{TO}_i, \tag{27}$$
$$\rho_{\text{stance}} = T_{\text{stance}}/T_{\text{stride}}, \tag{28}$$
$$\rho_{\text{swing}} = 1 - \rho_{\text{stance}}. \tag{29}$$

Cadence estimated from step times is

$$\text{Cadence} = \frac{60}{\overline{T}_{\text{step}}} \quad [\text{steps/min}]. \tag{30}$$

If bilateral events are reliable, double-support intervals may also be derived from overlapping stance intervals.

Spatial gait quantities

Do not report stride length or walking speed as primary validated outcomes unless the team implements and validates a positional reconstruction method. The present analysis can robustly support temporal and rotational/kinematic outcomes from synchronized IMUs; displacement derived by double integration requires a separate validation argument.

7.2 Joint-kinematic features

For joint angle $\theta_j^{(i)}(\tau)$:

$$\theta_{j,\max}^{(i)} = \max_{\tau} \theta_j^{(i)}(\tau), \tag{31}$$
$$\theta_{j,\min}^{(i)} = \min_{\tau} \theta_j^{(i)}(\tau), \tag{32}$$
$$\text{ROM}_j^{(i)} = \theta_{j,\max}^{(i)} - \theta_{j,\min}^{(i)}. \tag{33}$$

Peak timing is

$$\tau_{j,\max}^{(i)} = \arg\max_{\tau} \theta_j^{(i)}(\tau). \tag{34}$$

Compute these separately for hip, knee, and ankle on both sides. Phase-specific ROM can be calculated over stance and swing subintervals.

7.3 Variability and symmetry

For a positive scalar gait feature y measured over n cycles in one trial,

$$\text{CV}(y) = 100\frac{s_y}{\bar{y}}\ [\%]. \tag{35}$$

When outliers are expected, also report robust dispersion

$$\mathrm{rCV}(y) = 100\frac{1.4826\ \mathrm{MAD}(y)}{\mathrm{median}(y)}. \tag{36}$$

For bilateral feature values y_L and y_R, a bounded symmetry index is

$$\mathrm{SI}(y) = 100\frac{y_R - y_L}{\frac{1}{2}(|y_R| + |y_L|) + \epsilon}. \tag{37}$$

The raw left and right values must always be retained because no single symmetry index is universally optimal.

7.4 Trial-level classic feature vector

Let $\boldsymbol{g}_{scr} \in \mathbb{R}^{p_g}$ denote the final classic feature vector for trial r of participant s under condition c. It should contain only pre-specified, non-redundant outcomes, for example

$$\boldsymbol{g} = [T_{\mathrm{stride}}, T_{\mathrm{step}}, \rho_{\mathrm{stance}}, \mathrm{cadence}, \mathrm{ROM}_{LH}, \ldots, \mathrm{ROM}_{RA}, \mathrm{SI}_{\mathrm{stride}}, \mathrm{SI}_{\mathrm{ROM}}, \mathrm{CV}_{\mathrm{stride}}, \ldots]^\top. \tag{38}$$

A full feature dictionary is given in Appendix B.

8 Stage 1C: lower-limb multivariate functional PCA

8.1 Why multivariate rather than separate fPCA

If hip, knee, and ankle are decomposed independently, each fPCA explains variation of one joint only. To interpret components as *lower-limb functional movement modes* or synergies, the preferred formulation is multivariate functional PCA (MFPCA), in which all selected joint trajectories are components of one functional observation [7].

For each stride i, define the six-dimensional bilateral lower-limb function

$$\boldsymbol{X}_i(\tau) = \begin{bmatrix} X_{LH,i}(\tau) & X_{LK,i}(\tau) & X_{LA,i}(\tau) & X_{RH,i}(\tau) & X_{RK,i}(\tau) & X_{RA,i}(\tau) \end{bmatrix}^\top. \tag{39}$$

All trajectories use the same angle convention and normalized phase grid.

8.2 Weighted functional inner product

Knee motion can have larger variance than ankle or hip motion, so an unweighted decomposition can be dominated by the largest-amplitude joint. Define

$$\langle \boldsymbol{f}, \boldsymbol{g} \rangle_w = \sum_{p=1}^{P} w_p \int_0^1 f^{(p)}(\tau) g^{(p)}(\tau)\, d\tau, \tag{40}$$

with $P = 6$ and weights such as

$$w_p = \left[\int_0^1 \mathrm{Var}\{X^{(p)}(\tau)\}\, d\tau \right]^{-1}. \tag{41}$$

Weights may be normalized so $\sum_p w_p = P$. Equal weights should be retained as a sensitivity analysis.

8.3 Mean and covariance operator

The multivariate mean is

$$\boldsymbol{\mu}(\tau) = \frac{1}{N}\sum_{i=1}^{N} \boldsymbol{X}_i(\tau). \tag{42}$$

The cross-covariance function between channels p and q is

$$C_{pq}(s,t) = \frac{1}{N-1}\sum_{i=1}^{N}\left[X_i^{(p)}(s) - \mu^{(p)}(s)\right]\left[X_i^{(q)}(t) - \mu^{(q)}(t)\right]. \tag{43}$$

MFPCA eigenfunctions satisfy

$$\sum_{q=1}^{P} w_q \int_0^1 C_{pq}(s,t)\phi_k^{(q)}(t)\,dt = \lambda_k \phi_k^{(p)}(s), \tag{44}$$

with orthonormality

$$\langle \boldsymbol{\phi}_k, \boldsymbol{\phi}_l \rangle_w = \delta_{kl}. \tag{45}$$

The Karhunen–Loeve representation is

$$\boldsymbol{X}_i(\tau) = \boldsymbol{\mu}(\tau) + \sum_{k=1}^{\infty} \xi_{ik}\boldsymbol{\phi}_k(\tau), \tag{46}$$

where score

$$\xi_{ik} = \langle \boldsymbol{X}_i - \boldsymbol{\mu}, \boldsymbol{\phi}_k \rangle_w. \tag{47}$$

A K-component reconstruction is

$$\hat{\boldsymbol{X}}_i^{(K)}(\tau) = \boldsymbol{\mu}(\tau) + \sum_{k=1}^{K} \xi_{ik}\boldsymbol{\phi}_k(\tau). \tag{48}$$

8.4 Selecting the number of components

The fraction of variance explained by the first K modes is

$$\mathrm{FVE}(K) = \frac{\sum_{k=1}^{K}\lambda_k}{\sum_k \lambda_k}. \tag{49}$$

Use the smallest K satisfying a pre-specified threshold, e.g. 90–95% FVE, while also inspecting the scree plot and participant-level stability. A mode should not be interpreted biomechanically solely because it passes an FVE cutoff.

8.5 Balancing participants and trials

Because some participants contribute many more cycles than others, MFPCA must not be fit to a raw pool that allows one participant to dominate. Recommended options are:

1. draw the same maximum number of cycles per participant-condition-trial during model fitting; or
2. weight each cycle inversely by the number of cycles contributed by its trial/participant.

Participant-balanced resampling should be used to quantify mode stability.

8.6 Trial-level MFPCA features

For mode k, summarize cycle scores within each trial:

$$\bar{\xi}_{scrk} = \text{median}_i(\xi_{scrik}), \tag{50}$$

$$s_{\xi,scrk} = 1.4826\,\text{MAD}_i(\xi_{scrik}). \tag{51}$$

The trial-level MFPCA vector is

$$\boldsymbol{f}_{scr} = [\bar{\xi}_1, \ldots, \bar{\xi}_K, s_{\xi,1}, \ldots, s_{\xi,K}]^\top. \tag{52}$$

The primary phenotype vector may use only the mean/median scores if dispersion variables are highly redundant.

8.7 Comparing terrain and footwear in MFPCA space

For each retained mode k, compare participant-condition trial scores using a repeated-measures model when coverage allows:

$$\xi_{scrk} = \beta_{0k} + \boldsymbol{\beta}_k^\top \boldsymbol{C}_{sc} + b_{sk} + \epsilon_{scrk}, \qquad b_{sk} \sim \mathcal{N}(0, \sigma^2_{b,k}), \tag{53}$$

where $\boldsymbol{C}$ encodes terrain or footwear in the relevant matched sub-study. With very small matched participant counts, emphasize participant-linked effect plots, confidence intervals from participant-level resampling, and consistency of direction rather than a large set of p-values.

9 Stage 2A: whole-body transfer-entropy network

9.1 Primary network nodes

To match the planned directional block analysis, define a primary 12-node joint network

$$\mathcal{V}_{LL} = \{L_H, L_K, L_A, R_H, R_K, R_A\}, \tag{54}$$

$$\mathcal{V}_{UL} = \{L_S, L_E, L_W, R_S, R_E, R_W\}, \tag{55}$$

$$\mathcal{V} = \mathcal{V}_{LL} \cup \mathcal{V}_{UL}. \tag{56}$$

Here H, K, A, S, E, W denote hip, knee, ankle, shoulder, elbow, and wrist rotational signals respectively. These should be anatomically consistent scalar joint-angle trajectories or a pre-declared component of relative rotation, not RSS latent modes.

If reliable axial orientation is regenerated, pelvis/trunk variables may be added as a secondary set

$$\mathcal{V}_{AX} = \{P, T, \ldots\}. \tag{57}$$

The four requested primary blocks remain $LL \to LL$, $LL \to UL$, $UL \to UL$, and $UL \to LL$; axial-node results should be reported separately to avoid changing the main comparison structure.

9.2 TE analysis interval and equal information length

Transfer entropy is estimated from continuous time series in original sampling time, not from time-normalized cycle waveforms. Variable trial length creates estimator-bias differences, so each trial is converted to one or more fixed-information windows containing the same number M of complete accepted strides. A starting configuration is $M = 8$ strides, with sensitivity checks at 6 and 10 strides if data coverage permits.

If a trial contains $n > M$ valid strides, use either the central M strides or several non-overlapping M-stride windows and average trial-level network features. The choice must be fixed before outcome comparison.

9.3 Transfer entropy formulation

For source process X and target process Y, define history vectors

$$\boldsymbol{y}_t^{(k)} = (y_t, y_{t-1}, \ldots, y_{t-k+1}), \tag{58}$$

$$\boldsymbol{x}_{t-d}^{(l)} = (x_{t-d}, x_{t-d-1}, \ldots, x_{t-d-l+1}), \tag{59}$$

where d is the source-to-target delay. Schreiber transfer entropy is the conditional mutual information [8]

$$\mathrm{TE}_{X\to Y}(d) = I\left(Y_{t+1}; \boldsymbol{X}_{t-d}^{(l)} \mid \boldsymbol{Y}_t^{(k)}\right), \tag{60}$$

which can be written

$$\mathrm{TE}_{X\to Y}(d) = \sum p(y_{t+1}, \boldsymbol{y}_t^{(k)}, \boldsymbol{x}_{t-d}^{(l)}) \tag{61}$$

$$\times \log_2 \frac{p(y_{t+1} \mid \boldsymbol{y}_t^{(k)}, \boldsymbol{x}_{t-d}^{(l)})}{p(y_{t+1} \mid \boldsymbol{y}_t^{(k)})}. \tag{62}$$

The previous pilot study used $k = l = 1$, delays $d \in \{1, \ldots, 30\}$ samples, fixed one-degree bins, and selected the maximum TE across delays [3]. The full-body study retains $k = l = 1$ as the primary low-dimensional embedding unless sensitivity analysis demonstrates a clear need for larger histories.

9.4 Discretization

A direct fixed 1° binning scheme is easy to reproduce but can create sparse occupancy when applied to heterogeneous joints with different ranges. The recommended primary estimator uses a fixed number B of equal-frequency bins per signal within each TE window:

$$q_X(t) \in \{1, \ldots, B\}, \qquad P(q_X = b) \approx \frac{1}{B}. \tag{63}$$

A starting value such as $B = 8$ is computationally manageable and should be tested against $B \in \{6, 8, 10\}$. The original one-degree estimator should be retained as a sensitivity analysis for direct continuity with the pilot work.

Probability tables use a small pseudocount η only to prevent numerical zeros; the value of η must be negligible relative to observed counts and configuration-controlled.

9.5 Delay search and maximum-lag bias

Define the candidate delay set

$$\mathcal{D} = \{1, 2, \ldots, d_{\max}\}, \tag{64}$$

with $d_{\max} = 30$ samples (300 ms at 100 Hz) as the initial setting inherited from the pilot study. The observed edge statistic is

$$\mathrm{TE}_{X\to Y}^{\max} = \max_{d\in\mathcal{D}} \mathrm{TE}_{X\to Y}(d), \tag{65}$$

with selected lag

$$d_{XY}^{*} = \arg\max_{d\in\mathcal{D}} \mathrm{TE}_{X\to Y}(d). \tag{66}$$

Because taking a maximum over many lags creates upward selection bias, the same maximization must be applied to each null surrogate, not only to the observed pair.

9.6 Surrogate correction and edge significance

For edge $X \to Y$, generate B_s surrogate sources $X^{(b)}$ that preserve the source distribution and autocorrelation but destroy its directed alignment with Y. The default implementation uses random circular shifts constrained to be far from zero lag and, for strongly periodic gait, away from offsets near integer multiples of the median stride period. A subset of results should be checked with an alternative surrogate such as cycle-block permutation or phase-randomized surrogates.

For surrogate b compute

$$\mathrm{TE}_b^{\max} = \max_{d \in \mathcal{D}} \mathrm{TE}_{X^{(b)} \to Y}(d). \tag{67}$$

Define effective transfer entropy

$$\mathrm{ETE}_{X \to Y} = \mathrm{TE}_{X \to Y}^{\max} - \frac{1}{B_s} \sum_{b=1}^{B_s} \mathrm{TE}_b^{\max}. \tag{68}$$

An empirical one-sided p-value is

$$p_{XY} = \frac{1 + \sum_{b=1}^{B_s} \mathbb{I}(\mathrm{TE}_b^{\max} \geq \mathrm{TE}_{X \to Y}^{\max})}{B_s + 1}. \tag{69}$$

Use $B_s = 100$ during development and at least $B_s = 200$ for final results if runtime permits. Parallelize and cache surrogate computations.

Because many directed edges are tested, adjust p-values within each trial/network using the Benjamini–Hochberg false-discovery-rate procedure [9]. For target FDR $q = 0.05$, define

$$A_{ij} = \begin{cases} \max(0, \mathrm{ETE}_{i \to j}), & q_{ij} \leq 0.05, \\ 0, & \text{otherwise}, \end{cases} \qquad A_{ii} = 0. \tag{70}$$

The resulting $\boldsymbol{A} \in \mathbb{R}_+^{M \times M}$ is the directed TE adjacency matrix.

9.7 Interpretation: information transfer is not causal proof

Pairwise TE quantifies directed predictive information beyond the target's own past; it does not establish physiological causality. Shared gait phase, common neuromotor drive, and indirect paths can generate apparent directed edges. Journal language should therefore use *directed statistical coupling*, *directed predictive information*, or *information-transfer structure* rather than unconditional causal claims.

9.8 Optional phase-conditioned TE sensitivity analysis

To assess whether edges remain after controlling for common gait phase, define a phase variable Φ_t from the hybrid segmentation and compute

$$\mathrm{TE}_{X \to Y \mid \Phi}(d) = I\left(Y_{t+1}; \boldsymbol{X}_{t-d}^{(l)} \mid \boldsymbol{Y}_t^{(k)}, \Phi_t\right). \tag{71}$$

In discrete form,

$$\mathrm{TE}_{X \to Y \mid \Phi}(d) = \sum p(y_{t+1}, \boldsymbol{y}_t, \boldsymbol{x}_{t-d}, \phi_t) \tag{72}$$

$$\times \log_2 \frac{p(y_{t+1} \mid \boldsymbol{y}_t, \boldsymbol{x}_{t-d}, \phi_t)}{p(y_{t+1} \mid \boldsymbol{y}_t, \phi_t)}. \tag{73}$$

Because extra conditioning increases sample requirements, this is recommended as a robustness analysis using a small number of gait-phase bins rather than as the only primary TE estimator.

10 Stage 2B: TE network descriptors

10.1 Node-level strengths

For node i,

$$S_i^{\text{out}} = \sum_{j \neq i} A_{ij}, \tag{74}$$

$$S_i^{\text{in}} = \sum_{j \neq i} A_{ji}, \tag{75}$$

$$S_i^{\text{net}} = S_i^{\text{out}} - S_i^{\text{in}}. \tag{76}$$

Normalized fractions are

$$P_i^{\text{out}} = \frac{S_i^{\text{out}}}{\sum_k S_k^{\text{out}} + \epsilon}, \tag{77}$$

$$P_i^{\text{in}} = \frac{S_i^{\text{in}}}{\sum_k S_k^{\text{in}} + \epsilon}, \tag{78}$$

$$P_i^{\text{net}} = P_i^{\text{out}} - P_i^{\text{in}}. \tag{79}$$

These extend the inflow/outflow/cumulative concepts in the prior lower-limb pilot [3].

10.2 Block-level transfer

For node sets A and B, define average directed block transfer

$$\bar{T}_{A \to B} = \frac{1}{N_{AB}} \sum_{i \in A} \sum_{j \in B,\, j \neq i} A_{ij}, \tag{80}$$

where N_{AB} is the number of allowable directed edges. The four primary values are

$$\bar{T}_{LL \to LL}, \quad \bar{T}_{LL \to UL}, \quad \bar{T}_{UL \to UL}, \quad \bar{T}_{UL \to LL}. \tag{81}$$

An upper/lower directional balance is

$$B_{UL} = \frac{\bar{T}_{LL \to UL} - \bar{T}_{UL \to LL}}{\bar{T}_{LL \to UL} + \bar{T}_{UL \to LL} + \epsilon}. \tag{82}$$

10.3 Reciprocity and directional asymmetry

For node pair i, j, directional asymmetry is

$$D_{ij} = \frac{A_{ij} - A_{ji}}{A_{ij} + A_{ji} + \epsilon}. \tag{83}$$

A global weighted reciprocity descriptor is

$$R_w = \frac{2 \sum_{i<j} \min(A_{ij}, A_{ji})}{\sum_{i \neq j} A_{ij} + \epsilon}. \tag{84}$$

Network density after FDR thresholding is

$$\delta = \frac{\sum_{i \neq j} \mathbb{I}(A_{ij} > 0)}{M(M-1)}. \tag{85}$$

10.4 Anatomical aggregate descriptors

Additional interpretable aggregates should be defined before clustering, for example:

- ipsilateral versus contralateral transfer,
- left-to-right versus right-to-left transfer,
- proximal (hip/shoulder) versus distal (ankle/wrist) outflow,
- lower-body distal-to-proximal versus proximal-to-distal transfer,
- node-specific net flow for the 12 primary joints,
- significant-edge density and reciprocity.

The full 12-by-12 matrix should be saved and visualized, but clustering should use a reduced, pre-defined network feature vector $\boldsymbol{q} \in \mathbb{R}^{p_q}$ rather than all 132 raw directed edges by default.

11 Stage 2C: unified trial phenotype vector

For each valid trial, combine three feature families:

$$\boldsymbol{z}_{scr} = \begin{bmatrix} \boldsymbol{g}_{scr} \\ \boldsymbol{f}_{scr} \\ \boldsymbol{q}_{scr} \end{bmatrix} \in \mathbb{R}^{p_g+p_f+p_q}, \tag{86}$$

where

- $\boldsymbol{g}$ = classic gait features,
- $\boldsymbol{f}$ = lower-limb MFPCA scores,
- $\boldsymbol{q}$ = whole-body TE-network descriptors.

11.1 Robust scaling

For each scalar feature j across trials, use robust scaling

$$\tilde{z}_{ij} = \frac{z_{ij} - \text{median}(z_{\cdot j})}{1.4826\,\text{MAD}(z_{\cdot j}) + \epsilon}. \tag{87}$$

Features with near-zero variance are removed before similarity computation.

11.2 Block-balanced distance

Because the TE feature family may contain more variables than the classic or MFPCA families, simple Euclidean distance over all coordinates can overweight whichever block is largest. Define

$$d^2(i,j) = \alpha_g \frac{\|\tilde{\boldsymbol{g}}_i - \tilde{\boldsymbol{g}}_j\|_2^2}{p_g} + \alpha_f \frac{\|\tilde{\boldsymbol{f}}_i - \tilde{\boldsymbol{f}}_j\|_2^2}{p_f} + \alpha_q \frac{\|\tilde{\boldsymbol{q}}_i - \tilde{\boldsymbol{q}}_j\|_2^2}{p_q}, \tag{88}$$

with primary equal-family weighting

$$\alpha_g = \alpha_f = \alpha_q = \frac{1}{3}. \tag{89}$$

Sensitivity analysis should vary these weights or compare feature-family ablations.

11.3 Absolute and baseline-normalized phenotypes

The absolute phenotype is $\boldsymbol{z}_{scr}$. It answers: *what makes two complete gait trials similar?* It may naturally cluster by participant identity.

For adaptation analysis, define a participant-specific flat/reference baseline

$$\bar{\boldsymbol{z}}_{s0} = \frac{1}{R_{s0}} \sum_{r \in \mathcal{R}_{s0}} \boldsymbol{z}_{s0r} \tag{90}$$

and baseline change vector

$$\Delta \boldsymbol{z}_{scr} = \boldsymbol{z}_{scr} - \bar{\boldsymbol{z}}_{s0}. \tag{91}$$

This answers: *do different participants show similar changes away from their own normal gait?*

For robust implementation, the baseline should be calculated from the median across reference trials coordinate-wise rather than the arithmetic mean if outliers are present.

Primary clustering analyses

Run two distinct unsupervised analyses:

1. **Absolute phenotype clustering** on $\boldsymbol{z}$: reveals participant, terrain, footwear, or mixed structure.
2. **Adaptation phenotype clustering** on $\Delta \boldsymbol{z}$: removes much of each participant's baseline gait signature and targets recurring adaptation strategies.

Perform terrain-adaptation clustering and footwear-adaptation clustering on separate matched subsets to avoid condition confounding.

12 Stage 2D: similarity graph and spectral clustering

Spectral clustering is appropriate when clusters may be non-convex in the original feature space and can be represented through a similarity graph [10, 11].

12.1 Affinity matrix

Compute pairwise distances $d(i, j)$. Use a k_{NN}-nearest-neighbor graph to avoid a dense affinity matrix dominated by weak similarities. Let σ_i be the distance from point i to its k_{NN}th neighbor. Define self-tuning affinity

$$W_{ij} = \begin{cases} \exp\left[-\dfrac{d^2(i,j)}{\sigma_i \sigma_j + \epsilon}\right], & j \in \mathcal{N}_i \text{ or } i \in \mathcal{N}_j, \\ 0, & \text{otherwise}, \end{cases} \tag{92}$$

with $W_{ii} = 0$. A global RBF bandwidth based on the median non-zero pairwise distance should be retained as a sensitivity analysis.

12.2 Normalized graph Laplacian

Degree matrix:

$$D_{ii} = \sum_j W_{ij}. \tag{93}$$

Symmetric normalized Laplacian:

$$L_{\text{sym}} = I - D^{-1/2} W D^{-1/2}. \tag{94}$$

For candidate cluster count K_c, compute the eigenvectors $\boldsymbol{u}_1, \ldots, \boldsymbol{u}_{K_c}$ associated with the K_c smallest eigenvalues of L_{sym}. Form

$$U = [\boldsymbol{u}_1, \ldots, \boldsymbol{u}_{K_c}] \in \mathbb{R}^{N \times K_c} \tag{95}$$

and row-normalize

$$Y_{ij} = \frac{U_{ij}}{\sqrt{\sum_{m=1}^{K_c} U_{im}^2} + \epsilon}. \tag{96}$$

Apply K_c-means to rows of Y to obtain final cluster labels.

12.3 Selecting the number of clusters

Do not choose K_c solely because it best reproduces known terrain or subject labels. Use a combined unsupervised criterion:

- eigengap in L_{sym} spectrum,
- silhouette score in the spectral embedding,
- cluster-size reasonableness,
- stability under trial/participant bootstrap and feature-family perturbation.

A practical candidate range is $K_c = 2, \ldots, 6$ unless the number of usable trials is very small.

12.4 Post-hoc interpretation

Known metadata are used *after* clustering to characterize emergent clusters. Report contingency tables and association metrics such as adjusted Rand index (ARI) and normalized mutual information (NMI) between cluster labels and participant, terrain, or footwear labels. A strong match to participant identity in absolute space is not a failure; it quantifies individuality. Adaptation-space clusters are then examined to determine whether condition-specific or cross-condition strategies emerge.

For cluster c, compute feature centroid

$$\boldsymbol{\mu}_c = \frac{1}{|\mathcal{C}_c|} \sum_{i \in \mathcal{C}_c} \tilde{\boldsymbol{z}}_i \tag{97}$$

and feature contrast against the overall mean. Interpret clusters using classic gait, MFPCA, and TE features jointly rather than assigning biomechanical meaning from the spectral embedding alone.

13 Statistical analysis strategy

13.1 No pseudoreplication

Cycle-level values may be shown descriptively and used to estimate within-trial uncertainty, but population claims use trial or participant summaries. Bootstrap resampling for manuscript confidence intervals should resample participants first, then trials/cycles within participant where appropriate.

13.2 Terrain and footwear comparisons

For scalar trial feature y_{scr} in a matched sub-study, a simple mixed model is

$$y_{scr} = \beta_0 + \boldsymbol{\beta}^\top \boldsymbol{C}_{sc} + b_s + \epsilon_{scr}, \tag{98}$$

where b_s is a participant random intercept. With only a few participants sharing a comparison, report the model as estimation rather than as strong population inference. Include participant-linked plots and leave-one-participant-out sensitivity.

A baseline-relative feature can be defined

$$\Delta y_{scr} = y_{scr} - \text{median}_{r \in \mathcal{R}_{s0}} \, y_{s0r}. \tag{99}$$

The proportion of participants with the same effect direction is a useful descriptive robustness measure.

13.3 Multiple comparisons

Avoid testing every classic feature, MFPCA score, and TE edge with independent uncorrected p-values. Pre-specify primary outcomes. Use FDR within logical hypothesis families and report effect sizes with confidence intervals.

14 Sensitivity, ablation, and robustness analyses

The following analyses should be implemented as configuration-driven re-runs rather than manual one-off experiments.

Component	Required sensitivity analysis
Gait segmentation	foot vs shank event signal; gyro-only vs gyro+Hilbert fusion; event-confidence threshold; manual review subset.
MFPCA registration	event-registered shape normalization vs timing-preserving whole-stride normalization.
MFPCA scaling	inverse integrated-variance channel weights vs equal weights.
MFPCA model fit	participant-balanced subsampling; leave-one-participant-out mode stability.
TE discretization	equal-frequency $B = 6, 8, 10$; compare with original one-degree bins on a representative subset.
TE history	$k = l = 1$ primary; evaluate a small alternative history order if sample length supports it.
TE delay	$d_{\max} = 30$ samples primary; inspect lag distributions and repeat with shorter/longer range if needed.
TE null model	circular-shift surrogates; alternative cycle-block or phase-randomized surrogate on a subset.
TE common phase	compare pairwise effective TE with phase-conditioned TE sensitivity analysis.
Phenotype vector	classic-only, MFPCA-only, TE-only, classic+MFPCA, and all-feature ablations.
Similarity	local-scale kNN affinity vs global RBF bandwidth.
Clustering	candidate K_c values; bootstrap stability; participant leave-one-out.

15 Recommended manuscript result structure

If Stage 1 alone is used for a dataset-focused manuscript, recommended main outputs are:

1. dataset/cohort and sensor topology figure;
2. hybrid gait-segmentation diagram and event-validation examples;

3. classic gait parameter distributions by matched terrain/footwear;
4. lower-limb cycle-normalized trajectories;
5. MFPCA eigenfunctions, variance explained, and condition-specific score plots;
6. repeatability/QC table.

For the full Stage 1+Stage 2 manuscript, add:

1. representative whole-body TE adjacency matrices;
2. block transfer $LL \to LL$, $LL \to UL$, $UL \to UL$, $UL \to LL$ by condition;
3. node in/out/net-flow maps;
4. absolute and baseline-normalized spectral embeddings;
5. cluster metadata composition and cluster phenotype signatures;
6. feature-family ablation and clustering stability.

16 Codex-oriented software specification

16.1 Recommended project structure

```
Gait_Posture_Analysis/
  README.md
  pyproject.toml
  configs/
    default.yaml
    sensitivity/
      mfpca_equal_weights.yaml
      te_bins_6.yaml
      te_bins_10.yaml
      te_phase_conditioned.yaml
  src/gait_analysis/
    __init__.py
    paths.py
    metadata.py
    io.py
    qc.py
    orientation.py
    joint_angles.py
    gait_events.py
    hilbert_phase.py
    segmentation.py
    classical_features.py
    mfpca.py
    te_estimator.py
    te_surrogates.py
    te_network.py
    phenotype.py
    similarity.py
    spectral_clustering.py
    statistics.py
    visualization.py
    export.py
  notebooks/
    01_manifest_and_qc.ipynb
```

```
  02_segmentation_validation.ipynb
  03_stage1_classic_gait.ipynb
  04_stage1_mfpca.ipynb
  05_stage2_transfer_entropy.ipynb
  06_stage2_spectral_clustering.ipynb
  07_manuscript_figures.ipynb
tests/
  test_gait_events.py
  test_hilbert_phase.py
  test_segmentation.py
  test_classic_features.py
  test_mfpca.py
  test_te_estimator.py
  test_te_surrogates.py
  test_spectral_clustering.py
outputs/
  manifest/
  segmentation/
  stage1/
  stage2/
  figures/
  logs/
```

16.2 Core design rules for implementation

1. All parameters live in YAML/config objects. No analysis thresholds should be scattered as magic numbers in notebooks.
2. Every output table carries participant, condition, trial, source file, and configuration hash.
3. All random procedures use an explicit seed and save it in metadata.
4. Raw data are never modified in place.
5. Intermediate results are cached by source-file checksum plus configuration hash.
6. A failed recording produces a structured QC record rather than an exception that silently removes it from the cohort.
7. Notebooks call reusable package functions; notebooks do not contain the authoritative implementation.
8. Unit tests use synthetic signals with known event timing, known functional modes, known directionality, and known cluster structure.

16.3 Expected output tables

File	Minimum content
`cohort_manifest.csv`	participant, terrain, direction, footwear, heel code, trial, raw file, duration, sensors present, QC flags, inclusion status.
`gait_events.csv`	participant/trial, limb, event type, sample, time, gyro feature, Hilbert phase, confidence, accepted/rejected.
`gait_cycles.csv`	stride id, IC, TO, next IC, stance fraction, duration, phase-consistency error, QC.
`classic_cycle_features.csv`	one row per cycle with temporal, ROM, peak, symmetry components where defined.

File	Minimum content
`classic_trial_features.csv`	robust trial summaries used in phenotype vector.
`mfpca_model.npz`	phase grid, mean functions, eigenfunctions, eigenvalues, weights, configuration, training manifest hash.
`mfpca_cycle_scores.csv`	stride-level scores and reconstruction error.
`mfpca_trial_scores.csv`	median/dispersion of scores per trial.
`te_edges.csv`	trial/window, source, target, raw max TE, selected lag, surrogate mean, effective TE, p, q, significant flag.
`te_network_features.csv`	block flows, node strengths, net flows, reciprocity, density, laterality/proximal-distal descriptors.
`phenotype_absolute.csv`	scaled/unscaled complete trial phenotype vector.
`phenotype_delta.csv`	participant-baseline-normalized phenotype vectors.
`clusters.csv`	trial, representation type, graph config, cluster count, cluster label, spectral coordinates.
`cluster_stability.csv`	bootstrap run, ARI to reference solution, cluster sizes, feature/config perturbation.

16.4 Configuration schema

The following are starting defaults, not immutable scientific truths. Final values must be chosen from development data and sensitivity analyses.

```
data:
  sampling_rate_hz: 100
  primary_joint_angle_pipeline: validated_relative_rotation
  use_rss_modes_as_anatomical_angles: false

segmentation:
  preferred_event_sensor: foot
  fallback_event_sensor: shank
  filter_hz: [0.5, 6.0]
  hilbert_reference: shank_sagittal_gyro
  min_stride_s: 0.50
  max_stride_s: 2.00
  phase_cycle_tolerance: 0.20
  output_points_per_stride: 101

classic_features:
  include_stride_time: true
  include_step_time: true
  include_stance_swing: true
  include_cadence: true
  include_joint_rom: true
  include_peak_angles: true
  include_symmetry: true
  include_stride_length: false

mfpca:
  joints: [LH, LK, LA, RH, RK, RA]
  registration: event_registered
  channel_weighting: inverse_integrated_variance
  variance_explained_target: 0.95
  participant_balanced_fit: true

transfer_entropy:
  nodes: [LH, LK, LA, RH, RK, RA, LS, LE, LW, RS, RE, RW]
```

```
  history_target: 1
  history_source: 1
  delay_samples: [1, 30]
  discretization: quantile
  bins: 8
  te_window_strides: 8
  surrogate_method: circular_shift
  surrogates_development: 100
  surrogates_final: 200
  fdr_q: 0.05
  phase_conditioned_sensitivity: true

phenotype:
  scaler: robust_median_mad
  family_weights:
    classic: 0.333333
    mfpca: 0.333333
    transfer_entropy: 0.333333
  compute_absolute: true
  compute_subject_baseline_delta: true

spectral_clustering:
  affinity: local_scale_knn
  knn: 5
  candidate_clusters: [2, 3, 4, 5, 6]
  random_seed: 20260907
```

16.5 Algorithm 1: hybrid gait segmentation

```
function segment_trial(trial):
    load synchronized left/right foot and shank IMU signals
    load validated joint-angle signals
    run QC; abort with structured reason if required sensors are invalid

    for limb in [left, right]:
        event_signal = choose foot signal, else shank signal
        event_signal = zero_phase_filter(event_signal)
        event_signal = orient_sign_so_mid_swing_is_positive(event_signal)

        MS = detect_mid_swing_peaks(event_signal)
        for each MS_i:
            TO_i = detect_pre_mid_swing_terminal_contact(event_signal, MS_i)
            IC_i = detect_post_mid_swing_initial_contact(event_signal, MS_i)
            refine/check with foot stationarity and acceleration if available

        phi = hilbert_unwrapped_phase(reference_signal)
        fuse events with phase-cycle consistency
        reject impossible orderings / implausible durations / low confidence

    pair left/right events to obtain steps, strides, stance, swing
    save gait_events.csv and gait_cycles.csv
    return accepted cycles
```

16.6 Algorithm 2: Stage 1 classic features and MFPCA

```
for each accepted trial:
```

```
    for each accepted stride:
        compute stride/step/stance/swing features
        extract bilateral hip/knee/ankle trajectories
        save timing-preserving 0-100% trajectory
        save event-registered stance/swing trajectory
        compute cycle-level ROM, peaks, symmetry inputs

    robustly aggregate cycle features -> trial classic vector g

build participant-balanced MFPCA training set
estimate channel weights
fit multivariate mean/covariance/eigenfunctions
select K using target FVE + stability
project every accepted stride -> scores xi_1 ... xi_K
aggregate scores within each trial -> vector f
```

16.7 Algorithm 3: whole-body TE network

```
for each accepted trial:
    select equal-information TE windows (same number of complete strides)

    for each window:
        align 12 anatomically interpretable joint signals in original time
        discretize each signal with configured quantile bins

        for each directed pair X -> Y, X != Y:
            observed_curve = TE(X -> Y, delay=1..dmax)
            TE_obs = max(observed_curve)
            lag_star = argmax(observed_curve)

            for b in 1..B_surrogates:
                Xb = generate_null_surrogate(X)
                surrogate_curve = TE(Xb -> Y, delay=1..dmax)
                TE_null[b] = max(surrogate_curve)

            ETE = TE_obs - mean(TE_null)
            p = (1 + count(TE_null >= TE_obs)) / (B_surrogates + 1)

        apply Benjamini-Hochberg FDR to all directed edges in window
        construct thresholded adjacency A
        extract block and node network descriptors

    aggregate window descriptors -> trial TE vector q
```

16.8 Algorithm 4: phenotype construction and spectral clustering

```
for each trial with complete Stage-1 and Stage-2 outputs:
    z = concatenate(classic_vector g, mfpca_vector f, te_vector q)

robust-scale each feature across eligible trials
compute family-balanced pairwise distances

absolute analysis:
    W_abs = local_scale_knn_affinity(z)
    run spectral clustering for K in candidate range
```

```
adaptation analysis:
    for each participant:
        baseline = robust_center(reference flat/reference-shoe trials)
        delta_z = z - baseline
    run terrain and footwear subsets separately
    W_delta = local_scale_knn_affinity(delta_z)
    run spectral clustering for K in candidate range

select K using eigengap + silhouette + bootstrap stability
interpret clusters post-hoc using participant/terrain/footwear metadata
```

17 Validation and test plan for the software

17.1 Segmentation tests

Generate synthetic quasi-periodic signals with known IC/TO/MS timestamps and perturb them with noise, drift, amplitude changes, missing peaks, and variable stride duration. Assert:

- event ordering is preserved;
- event timing error stays within a configurable tolerance;
- missing events produce a QC flag rather than fabricated events;
- left/right alternation rules reject impossible sequences;
- the Hilbert phase advances by approximately one cycle per accepted stride.

A manually reviewed subset of real RMD trials should be used as a regression test set.

17.2 MFPCA tests

Create synthetic multivariate functions

$$\boldsymbol{X}_i(\tau) = \boldsymbol{\mu}(\tau) + a_i\boldsymbol{\phi}_1(\tau) + b_i\boldsymbol{\phi}_2(\tau) + \boldsymbol{\epsilon}_i(\tau) \tag{100}$$

with known orthogonal modes. Assert recovery of the subspace, descending eigenvalues, orthonormality, and low reconstruction error.

17.3 TE tests

Use simulations with known directionality, e.g.

$$X_t = 0.8X_{t-1} + \epsilon_t^X, \tag{101}$$

$$Y_t = 0.6Y_{t-1} + 0.5X_{t-d} + \epsilon_t^Y. \tag{102}$$

Expected result:

$$\mathrm{ETE}_{X\to Y} > \mathrm{ETE}_{Y\to X} \tag{103}$$

and selected lag should concentrate near the simulated d. Independent processes should rarely survive FDR after surrogate correction.

17.4 Spectral-clustering tests

Construct synthetic phenotype vectors with non-convex or graph-separable clusters and verify that the normalized Laplacian implementation reproduces known labels up to permutation. Test disconnected graphs, isolated points, duplicated trials, and zero-variance features.

18 Failure modes and decision rules

Failure mode	Required response
No reliable foot signal	fall back to shank gyro; reduce event confidence; record sensor used.
Foot and shank disagree strongly	retain neither automatically; mark event window for review or exclude affected cycles.
Hilbert phase is non-monotonic / multiple rotations per stride	inspect filtering/reference signal; reject cycle if phase consistency cannot be recovered.
Too few valid cycles	exclude from MFPCA/TE according to pre-specified minimum; retain in manifest.
Missing upper-limb joint	do not impute TE node from another trial; either use a declared reduced network or exclude that trial from whole-body TE.
TE window shorter than required	exclude Stage 2 for that trial or use a pre-declared shorter sensitivity configuration; do not silently pad data.
TE adjacency becomes nearly empty	inspect bin count, sample size, surrogate test, and FDR; report sparsity rather than lowering significance threshold after seeing results.
TE adjacency becomes nearly complete	inspect common-phase confounding and estimator bias; run phase-conditioned/surrogate sensitivity analysis.
MFPCA component dominated by one participant	rebalance fit and quantify leave-one-participant-out stability.
Spectral graph disconnected unexpectedly	inspect scaling, kNN choice, missing data, and duplicate vectors; do not force a visually pleasing clustering.
Clusters reproduce participant identity only	report this for absolute phenotype, then inspect baseline-normalized adaptation clustering.
Terrain and shoe effects confounded	split into matched terrain/reference-shoe and flat-terrain/footwear analyses.

19 Recommended final analysis sequence

The following sequence is the recommended locked order for the project:

1. Freeze the cohort manifest and QC rules.
2. Lock the anatomically interpretable joint-angle pipeline.
3. Implement and validate hybrid foot/shank-gyro + Hilbert segmentation.
4. Generate cycle and trial classic gait features.
5. Fit participant-balanced lower-limb MFPCA and derive trial scores.
6. Complete Stage 1 terrain and footwear analyses before viewing Stage 2 clusters.
7. Implement surrogate-corrected whole-body TE on equal-information windows.
8. Reduce the TE matrix to pre-defined node/block network descriptors.
9. Construct absolute and participant-baseline-normalized phenotype vectors.
10. Run spectral clustering separately for terrain and footwear matched subsets.

11. Quantify cluster stability and feature-family ablations.

12. Only then finalize biomechanical interpretation and manuscript claims.

This ordering reduces analytic flexibility and makes the final manuscript easier to defend because each derived layer can be validated before it feeds the next one.

A Compact mathematical reference

A.1 Hilbert phase

$$\hat{x}(t) = \mathcal{H}\{x(t)\}, \tag{104}$$
$$z(t) = x(t) + j\hat{x}(t), \tag{105}$$
$$\phi(t) = \arg z(t), \tag{106}$$
$$\Phi(t) = (\phi(t) \bmod 2\pi)/(2\pi). \tag{107}$$

A.2 MFPCA

$$\boldsymbol{X}_i(\tau) = \boldsymbol{\mu}(\tau) + \sum_k \xi_{ik}\boldsymbol{\phi}_k(\tau), \tag{108}$$
$$\xi_{ik} = \sum_p w_p \int_0^1 (X_i^{(p)} - \mu^{(p)})\phi_k^{(p)}\, d\tau, \tag{109}$$
$$\mathrm{FVE}(K) = \frac{\sum_{k=1}^{K}\lambda_k}{\sum_k \lambda_k}. \tag{110}$$

A.3 Transfer entropy

$$\mathrm{TE}_{X\to Y}(d) = I(Y_{t+1}; X_{t-d}^{(l)} \mid Y_t^{(k)}), \tag{111}$$
$$\mathrm{ETE}_{X\to Y} = \max_d \mathrm{TE}_{X\to Y}(d) - \mathbb{E}_{null}\left[\max_d \mathrm{TE}_{X^{null}\to Y}(d)\right]. \tag{112}$$

A.4 Spectral clustering

$$D_{ii} = \sum_j W_{ij}, \tag{113}$$
$$L_{sym} = I - D^{-1/2}WD^{-1/2}. \tag{114}$$

Use the K_c smallest eigenvectors, row-normalize, and apply K_c-means.

B Feature dictionary

Family	Feature	Definition / role
Classic	stride time	same-foot IC-to-IC interval.
Classic	step time	alternating contralateral IC interval.

Family	Feature	Definition / role
Classic	stance time/fraction	IC-to-TO duration and normalized fraction.
Classic	swing time/fraction	TO-to-next-IC duration and normalized fraction.
Classic	cadence	steps per minute from step intervals.
Classic	joint ROM	max minus min of hip/knee/ankle angle per cycle.
Classic	peak angle/timing	extremum and its normalized gait phase.
Classic	temporal variability	CV/rCV of stride, step, stance, swing.
Classic	bilateral symmetry	bounded normalized left-right difference plus raw bilateral values.
MFPCA	score ξ_k	projection onto functional mode k.
MFPCA	score dispersion	within-trial variability of mode score.
MFPCA	reconstruction error	residual functional energy after retained modes.
TE network	node out-strength	sum of significant effective outgoing TE.
TE network	node in-strength	sum of significant effective incoming TE.
TE network	node net flow	out-strength minus in-strength.
TE network	$LL \rightarrow LL$	mean directed transfer within lower-limb nodes.
TE network	$LL \rightarrow UL$	mean lower-to-upper directed transfer.
TE network	$UL \rightarrow UL$	mean directed transfer within upper-limb nodes.
TE network	$UL \rightarrow LL$	mean upper-to-lower directed transfer.
TE network	upper/lower balance	normalized difference of $LL \rightarrow UL$ and $UL \rightarrow LL$.
TE network	reciprocity	fraction of weighted bidirectional overlap.
TE network	density	fraction of FDR-significant directed edges.
TE network	laterality	ipsilateral/contralateral and left/right aggregate transfer.

C Suggested manuscript hypotheses

A compact hypothesis set for the combined Scientific Reports route is:

1. Terrain and footwear perturbations produce reproducible within-participant changes in classic gait timing and lower-limb kinematics relative to reference walking.
2. These perturbations alter the scores of a low-dimensional set of bilateral lower-limb functional movement modes obtained by MFPCA.
3. Whole-body directed information-transfer networks reorganize across terrain and footwear, including changes in lower-to-lower, lower-to-upper, upper-to-upper, and upper-to-lower transfer structure.
4. Trial phenotypes combining classic gait, functional-mode scores, and TE-network descriptors form stable similarity structures that reflect both participant-specific gait signatures and recurring baseline-normalized adaptation strategies.

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